

# BY SIR OTONDE MICHAEL AWITI-DELEGATE

## AGE 110 PAST PAPER WITH ANSWERS

### BY SIR OTONDE MICHAEL AWITI

#### Q.1a) Define cartography (3 marks)

**Cartography** is the **science, art, and technique of map making**, which involves the **collection, processing, design, and presentation of geographical data** to communicate spatial information effectively.

#### b) Discuss the attributes of good craftsmanship (3 marks)

Good cartographic craftsmanship is characterized by:

- **Accuracy** – correct representation of spatial data.
- **Clarity and legibility** – symbols, lettering, and colors are easy to read.
- **Aesthetic balance** – proper layout, neatness, and visual harmony of map elements.

#### c) What is plottable error in cartography and how useful is it? (3 marks)

**Plottable error** is the **smallest distance that can be accurately plotted or measured on a map**, usually about **0.25 mm**.

It is useful in:

- Determining the **accuracy limits of a map**.
- Selecting an **appropriate map scale**.
- Avoiding misinterpretation of very small features.

#### d) Explain the three pertinent stages in the study of maps (3 marks)

The three stages are:

1. **Map compilation** – collection and selection of geographical data.
2. **Map construction** – symbolization, generalization, and drawing of the map.
3. **Map interpretation** – reading, analysis, and extraction of information from the map.

#### e) State the main marginal information on topographical maps and show their usefulness (3 marks)

Main marginal information includes:

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- **Scale** – helps measure distance.
- **Legend (key)** – explains symbols used.
- **Direction (north arrow)** – aids orientation.
- **Contour interval and projection** – assist in relief interpretation and accuracy.

**f) Define relief and distinguish the qualitative and quantitative methods used to depict relief on maps (3 marks)**

**Relief** refers to the **variation in elevation and slope of the Earth's surface**.

- **Qualitative methods** show relief without exact heights (e.g. hachures, shading).
- **Quantitative methods** show relief with numerical values (e.g. contours, spot heights, trigonometrical stations).

**g) Explain the concept of stream ordering as proposed by Horton (1945) (3 marks)**

**Stream ordering** is the classification of streams based on their hierarchy. According to **Horton (1945)**:

- The smallest unbranched streams are **first-order**.
- When two streams of the same order join, they form a **higher-order stream**.
- Stream order increases downstream.

**h) Define drainage density and show its usefulness in geomorphological studies (3 marks)**

**Drainage density** is the **total length of all streams in a drainage basin divided by the basin area**.

It is useful in:

- Assessing **surface runoff and infiltration**.
- Understanding **rock type and permeability**.
- Interpreting **erosional characteristics of landscapes**.

**i) Briefly explain the procedure followed in network analysis in economic geography (3 marks)**

The procedure involves:

1. **Identification of nodes** (centers such as towns or markets).
2. **Establishment of links** (transport or communication routes).

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3. **Analysis of flow, cost, and accessibility** to determine efficiency and connectivity.

**j) Discuss the usefulness of maps in the modern era (3 marks)**

Maps are useful in the modern era for:

- **Navigation and transportation** (GPS, digital maps).
- **Urban planning and resource management.**
- **Disaster management, environmental monitoring, and decision-making** using GIS and remote sensing.

**Q.2: Discuss the history of the development of cartography highlighting its key areas (15 marks)**

**1. Prehistoric cartography**

Early humans produced crude maps on cave walls, rocks, and animal skins. These maps showed hunting routes, water sources, and shelters, proving that spatial awareness existed long before writing.

**2. Mesopotamian cartography**

Babylonians drew maps on clay tablets showing towns, rivers, and boundaries. These maps introduced ideas of **scale, symbols, and orientation**, laying foundations for formal cartography.

**3. Egyptian cartography**

Maps were used for land measurement, taxation, and irrigation planning along the Nile. Egyptians developed **surveying techniques** after floods erased boundaries.

**4. Greek cartography**

Greeks transformed cartography into a science by using mathematics and astronomy. They believed the Earth was spherical and attempted accurate world representations.

**5. Eratosthenes' contribution**

Eratosthenes calculated the **circumference of the Earth** with remarkable accuracy. This was crucial for understanding scale and global distance measurement.

**6. Ptolemaic cartography**

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Ptolemy introduced **latitude and longitude** and map projections. His book *Geographia* influenced map-making for over 1,000 years.

## 7. Roman cartography

Romans focused on **practical maps** for roads, military movement, and administration. Accuracy was less important than usability.

## 8. Medieval cartography

Maps were dominated by religion rather than science. **T-O maps** placed Jerusalem at the center, showing symbolic rather than spatial accuracy.

## 9. Islamic cartography

Islamic scholars improved accuracy using astronomy and travelers' reports. **Al-Idrisi** produced one of the most accurate medieval world maps.

## 10. Renaissance cartography

Rediscovery of classical knowledge led to scientific mapping. The **printing press** allowed mass production and standardization of maps.

## 11. Age of Exploration

European exploration demanded better nautical charts. **Portolan maps** were developed for navigation using compass directions.

## 12. 18th–19th century cartography

Triangulation, precise surveying, and national mapping agencies improved accuracy and consistency in topographical maps.

## 13. 20th century cartography

Aerial photography and photogrammetry revolutionized map accuracy, especially for inaccessible areas.

## 14. Modern cartography

Computers introduced GIS, GPS, and remote sensing, allowing data storage, analysis, and real-time mapping.

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## **15. Digital and web cartography**

Online interactive maps enable instant updates, user interaction, and real-time navigation, marking the digital era of cartography.

### **Q.3: Describe the main instruments and tools used for drawing maps (15 marks)**

#### **1. Drawing board**

Provides a stable, flat surface to prevent distortion while drawing maps.

#### **2. Paper and tracing paper**

Used for drafting and copying maps while preserving accuracy and neatness.

#### **3. Pencils**

Different grades allow light sketching and precise detailing before final inking.

#### **4. Eraser**

Used to correct mistakes and refine map details without damaging the paper.

#### **5. Ruler / straight edge**

Ensures accurate straight lines for grids, borders, and scale bars.

#### **6. Scale ruler**

Allows correct conversion of ground distances to map distances.

#### **7. Compasses**

Used to draw arcs, circles, and measure distances accurately.

#### **8. Protractor**

Used to measure and plot angles and bearings on maps.

#### **9. French curves**

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Assist in drawing smooth curves such as rivers, roads, and contours.

### **10. Templates**

Provide uniform symbols for settlements, airports, and other features.

### **11. Technical pens**

Ensure uniform ink thickness and professional map presentation.

### **12. Lettering stencils**

Used for neat, consistent, and legible map lettering.

### **13. Color pencils and inks**

Differentiate map features and improve visual communication.

### **14. Planimeter**

Measures areas of irregular shapes accurately on maps.

### **15. Modern digital tools**

GIS software, scanners, digitizers, and plotters allow fast, accurate digital map production.

## **Q.4: Discuss the nearest neighbour technique in settlement geography (15 marks)**

### **1. Meaning**

It is a quantitative method used to analyze the spatial distribution of settlements.

### **2. Origin**

Developed by **Clark and Evans (1954)** to study point patterns statistically.

### **3. Purpose**

Helps determine whether settlements are clustered, random, or regularly spaced.

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**4. Point-based analysis**

Each settlement is treated as a point on the map.

**5. Distance measurement**

The distance from each settlement to its nearest neighbor is measured.

**6. Observed mean distance**

The average of all nearest-neighbour distances calculated from the map.

**7. Expected mean distance**

The theoretical distance assuming random distribution.

**8. Nearest Neighbour Index (R)**

Calculated by dividing observed distance by expected distance.

**9. Interpretation of  $R = 1$**

Indicates random settlement distribution.

**10. Interpretation of  $R < 1$**

Indicates clustered or nucleated settlements.

**11. Interpretation of  $R > 1$**

Indicates dispersed or regular settlement patterns.

**12. Application**

Used in rural settlement studies and urban planning.

**13. Strength**

Provides objective and measurable results.

**14. Limitation**

Assumes a uniform landscape and ignores physical barriers.

## **15. Importance**

Despite limitations, it is widely used in settlement geography.

## **Q.5: Discuss factors influencing drainage density (15 marks)**

### **1. Rock type**

Impermeable rocks promote runoff, increasing drainage density.

### **2. Soil permeability**

Permeable soils allow infiltration, reducing surface streams.

### **3. Rainfall amount**

High rainfall supplies more water for stream formation.

### **4. Rainfall intensity**

Heavy rainfall causes rapid surface runoff.

### **5. Relief**

Steep slopes speed runoff, increasing channel development.

### **6. Vegetation cover**

Dense vegetation reduces runoff and erosion.

### **7. Climate**

Humid climates favor dense drainage networks.

### **8. Basin size**

Smaller basins often have higher drainage density.

### **9. Geological structure**

Faults and joints guide stream flow paths.

#### **10. Land use**

Urban areas increase impervious surfaces and runoff.

#### **11. Human activities**

Deforestation increases erosion and stream formation.

#### **12. Stage of erosion**

Young landscapes have more active stream development.

#### **13. Surface roughness**

Smooth surfaces increase overland flow.

#### **14. Evaporation rate**

High evaporation reduces available surface water.

#### **15. Time**

Long-term erosion modifies drainage patterns.

### **Q1**

#### **a) Briefly explain how cartography evolved. (5 marks)**

Cartography evolved through several stages. Early humans produced simple sketch maps to show hunting grounds and settlements. Ancient civilizations such as the Egyptians and Babylonians used maps for land measurement and administration. The Greeks introduced scientific cartography through mathematics and astronomy, with Ptolemy developing latitude and longitude. During the Renaissance and Age of Exploration, improved navigation led to accurate maps and projections. In the modern era, cartography has advanced through aerial photography, GIS, GPS, and digital mapping technologies.

#### **b) Define plottable error in map drawing. (5 marks)**

Plottable error is the **smallest distance that can be accurately plotted or measured on a map**, usually taken as **0.25 mm**. It represents the limit of human visual accuracy when drawing or reading maps. Plottable error determines the **minimum size of features** that can be shown on a

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map. It helps cartographers choose appropriate map scales. It also prevents overloading maps with excessive detail that cannot be accurately represented.

## c) What is the usefulness of maps? (5 marks)

Maps are useful for:

1. **Locating places and features** accurately on the Earth's surface.
2. **Navigation and route planning** on land, sea, and air.
3. **Planning and decision-making** in urban development and resource management.
4. **Analyzing spatial patterns** such as population, climate, and land use.
5. **Education and communication** of geographical information clearly and effectively.

## d) Explain the major marginal information on maps. (5 marks)

Major marginal information includes:

1. **Title** – shows the subject and area covered by the map.
2. **Scale** – allows measurement of ground distances.
3. **Legend (key)** – explains symbols and signs used.
4. **Direction (north arrow)** – helps orientation.
5. **Contour interval, projection, and source** – provide accuracy, relief interpretation, and reliability of the map.

## e) Distinguish qualitative and quantitative mapping. (5 marks)

**Qualitative mapping** shows the **type or presence of features** without numerical values, using symbols or colors (e.g. vegetation types).

**Quantitative mapping** represents **numerical data**, such as population density or rainfall amounts.

Qualitative maps emphasize **description**, while quantitative maps emphasize **measurement**.

Qualitative maps are often categorical, while quantitative maps are statistical.

Examples include land-use maps (qualitative) and choropleth maps (quantitative).

## f) Differentiate drainage density and stream intensity. (5 marks)

**Drainage density** is the **total length of streams per unit area of a drainage basin**.

**Stream intensity** (stream frequency) is the **number of streams per unit area**.

Drainage density emphasizes **length of channels**, while stream intensity emphasizes **number of channels**.

Drainage density reflects runoff and permeability.

Stream intensity reflects degree of dissection of the landscape.

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## **Q2**

**‘Cartography is an art, science and technology of map making’ — Discuss. (20 marks)**

Cartography is regarded as an **art, science, and technology** because it combines creativity, accuracy, and modern tools.

As an **art**, cartography involves aesthetic design, symbol selection, color harmony, and balanced layout to ensure maps are visually appealing and easy to read.

As a **science**, it relies on mathematics, geometry, geodesy, and statistics to ensure accuracy in scale, projection, distance, and direction.

Cartography applies scientific principles in data collection, classification, and generalization.

As a **technology**, cartography uses modern tools such as computers, GIS, GPS, remote sensing, and digital mapping software.

Technological advances allow real-time data processing, spatial analysis, and interactive maps.

Cartography integrates data from satellites and aerial photography.

It supports thematic mapping for planning and research.

It enables automation and mass production of maps.

Modern cartography supports decision-making and modeling.

Therefore, cartography is a multidisciplinary field combining **artistic design, scientific accuracy, and technological innovation**

**Q3 What is map draftsmanship? Discuss its qualities. (20 marks)**

**Map draftsmanship** refers to the **skill, technique, and artistic ability involved in drawing maps neatly, accurately, and clearly**, so that geographical information is communicated effectively to the map user. It combines accuracy, clarity, aesthetics, and technical competence in map production. Good map draftsmanship ensures that a map is not only correct but also easy to read, interpret, and use.

**Qualities of good map draftsmanship**

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1. **Accuracy in representation of features**  
All geographical features such as rivers, roads, settlements, and boundaries must be correctly positioned and proportioned according to the map scale. Inaccurate representation leads to misleading interpretation.
2. **Neatness and cleanliness**  
A well-drafted map should be free from smudges, unnecessary marks, and uneven lines. Neat maps enhance readability and create a professional appearance.
3. **Proper symbolization**  
Symbols used must be standardized, simple, and appropriate for the features they represent. Correct symbolization avoids confusion and ensures uniform interpretation.
4. **Clear and legible lettering**  
Lettering should be uniform, well-spaced, and readable. Names of places, rivers, and other features must not overlap or obscure important map details.
5. **Balanced layout and spacing**  
Map elements such as the title, legend, scale, and north arrow should be well arranged to avoid congestion in one area and emptiness in another.
6. **Appropriate scale usage**  
The chosen scale should suit the purpose of the map. Features must be drawn according to scale to maintain correct size relationships.
7. **Consistent line thickness**  
Lines should vary in thickness according to importance but remain consistent for similar features. For example, main roads should be thicker than footpaths.
8. **Correct use of colors**  
Colors should be used sparingly and meaningfully to distinguish features. Overuse of color can confuse users, while correct use improves visual appeal.
9. **Simplicity and clarity**  
A good map avoids unnecessary detail. Only essential information should be included to ensure the message of the map is clear.
10. **Proper marginal information**  
All essential marginal details such as title, scale, legend, source, projection, and date should be included and clearly presented.
11. **Visual balance and harmony**  
Map elements must relate well visually. There should be harmony between symbols, colors, lettering, and empty spaces.
12. **Avoidance of overcrowding**  
Features should not overlap excessively. Overcrowding reduces readability and makes interpretation difficult.
13. **Correct orientation**  
The map must be properly oriented, usually with north at the top, unless otherwise stated. This ensures consistency and ease of use.
14. **Precision in measurements**  
Distances, angles, and areas must be accurately measured and plotted to maintain map reliability.

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**15. Professional finish**

The final map should appear complete, polished, and well-presented, suitable for academic, planning, or professional use.

**Conclusion**

Good map draftsmanship is essential for effective cartographic communication. It ensures that maps are **accurate, readable, visually appealing, and reliable**, enabling users to interpret geographical information correctly. Without good draftsmanship, even accurate data may fail to communicate its intended message.

**Q4 Discuss the stages involved in the study of maps. (20 marks)**

The study of maps involves a systematic process that begins with **map preparation** and ends with **map use and interpretation**. These stages ensure that maps are accurate, clear, and useful for geographical analysis and decision-making.

**1. Data collection and compilation**

This is the initial stage where geographical data is gathered from sources such as field surveys, aerial photographs, satellite imagery, census records, and existing maps. The accuracy of a map largely depends on the quality of data collected.

**2. Selection and classification of data**

Not all collected data can be shown on a single map. Relevant data is selected based on the map's purpose and classified into categories such as physical, cultural, or economic features to simplify representation.

**3. Choice of map projection**

Since the Earth is spherical, a projection must be chosen to represent it on a flat surface. The projection selected depends on whether the map emphasizes area, shape, distance, or direction.

**4. Scale selection**

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Scale determines the relationship between distances on the map and the ground. A suitable scale is chosen depending on the area covered and the level of detail required.

## 5. Generalization of features

Generalization involves simplifying, exaggerating, or omitting details that cannot be clearly shown at the chosen scale. This prevents overcrowding and improves map readability.

## 6. Symbolization

Symbols, colors, and patterns are assigned to represent different features. Effective symbolization ensures that information is easily understood by the map user.

## 7. Map construction

At this stage, features are accurately drawn on the map according to scale, projection, and symbols. This forms the basic structure of the map.

## 8. Lettering and layout

Place names and labels are added carefully to avoid overlap. Map elements such as title, legend, scale, and north arrow are arranged in a balanced manner.

## 9. Coloring and finishing

Color is applied to enhance visual appeal and differentiate features. Finishing includes inking, cleaning, and final corrections to improve presentation.

## 10. Map reproduction

The completed map is reproduced through printing or digital duplication. This allows maps to be distributed to users without loss of quality.

## 11. Map reading

This stage involves identifying symbols, understanding scale, direction, and recognizing map features. It is the first step in map use.

## 12. Map interpretation

Interpretation goes beyond reading to explain spatial relationships and patterns such as settlement distribution, relief, or drainage systems.

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### **13. Measurement and analysis**

Measurements of distance, area, gradient, and direction are made using map scales and tools. These measurements support quantitative analysis.

### **14. Comparison with other maps**

Maps may be compared with others of different periods or themes to identify changes, trends, and spatial variations.

### **15. Drawing conclusions from map data**

The final stage involves summarizing findings and making inferences for planning, research, or decision-making based on map analysis.

### **Conclusion**

The stages involved in the study of maps ensure that maps are **accurately produced, clearly presented, and effectively used**. Each stage plays a vital role in transforming raw geographical data into meaningful spatial information.

## **Q5. Discuss the nearest neighbour technique as a method of point pattern analysis in settlement geography. (20 marks)**

The **nearest neighbour technique** is a **quantitative statistical method** used in settlement geography to analyze the **spatial distribution of settlements**. It helps determine whether settlements are **clustered, randomly distributed, or regularly spaced** within a given area.

### **1. Meaning of the nearest neighbour technique**

The technique examines the spatial relationship between settlements by measuring the distance from each settlement to its closest neighboring settlement.

### **2. Origin of the technique**

The method was developed by **Clark and Evans in 1954** to provide an objective way of analyzing point patterns rather than relying on visual observation alone.

### **3. Settlements treated as points**

Each settlement, regardless of its size or population, is represented as a single point on the map to simplify analysis.

## 4. Measurement of nearest neighbour distances

For every settlement, the distance to the closest neighbouring settlement is measured using map scales or digital tools.

## 5. Calculation of observed mean distance

All measured nearest-neighbour distances are added together and divided by the number of settlements to obtain the **observed mean distance**.

## 6. Calculation of expected mean distance

An expected mean distance is calculated assuming settlements are randomly distributed across the area, using a mathematical formula.

## 7. Nearest Neighbour Index (R)

The **Nearest Neighbour Index (R)** is obtained by dividing the observed mean distance by the expected mean distance.

## 8. Interpretation of $R = 1$

An R-value of **1** indicates a **random settlement pattern**, where settlements are neither clustered nor regularly spaced.

## 9. Interpretation of $R < 1$

An R-value **less than 1** indicates a **clustered or nucleated pattern**, often caused by factors such as water sources, fertile land, or social ties.

## 10. Interpretation of $R > 1$

An R-value **greater than 1** indicates a **regular or dispersed settlement pattern**, often found in planned settlements or areas with evenly distributed resources.

## 11. Explanation of settlement spacing

The technique helps explain why settlements are spaced the way they are in relation to physical, economic, and social factors.

## 12. Application in rural settlement studies

In rural areas, it helps analyze village spacing and land-use patterns such as farming systems.

### **13. Application in urban studies**

In urban geography, the technique is used to study the distribution of facilities such as schools, hospitals, and markets.

### **14. Role in planning and resource allocation**

Planners use nearest neighbour analysis to identify areas that are underserved or overconcentrated with services.

### **15. Objectivity of the method**

The technique provides **objective and measurable results**, reducing personal bias in spatial analysis.

### **16. Statistical reliability**

Because it is based on numerical calculations, the method allows comparisons between different regions and time periods.

### **17. Limitation: assumption of uniform terrain**

The technique assumes a uniform, flat surface and ignores physical barriers such as mountains, rivers, and lakes.

### **18. Limitation: ignores settlement size**

All settlements are treated equally, regardless of population or economic importance.

### **19. Limitation: boundary effects**

Settlements near the edge of the study area may distort results because neighbouring settlements outside the boundary are ignored.

### **20. Importance despite limitations**

Despite its limitations, the nearest neighbour technique remains widely used because of its simplicity, clarity, and effectiveness in point pattern analysis.

### **Conclusion**

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The nearest neighbour technique is an important quantitative tool in settlement geography. It provides a clear and objective method for analyzing settlement patterns and contributes significantly to planning, research, and spatial decision-making.



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